# Cryptography Exercises

**under construction**
**do not disitribute**

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## 1 Statistical distance and information-theoretic encryption

Let $A$ and $B$ be two random variables taking values in a finite nonempty set $\mathcal{X}$. Define the *statistical distance*[1] *between $A$ and $B$* as

$$\Delta(A\,;B) := \tfrac{1}{2} \cdot \sum_{a \in \mathcal{X}} |\Pr[A=a] - \Pr[B=a]|,$$

and *statistical distance of $A$ from uniformity* as

$$d(A) := \Delta(A\,;U_{\mathcal{X}}),$$

where $U_{\mathcal{X}}$ is uniform over $\mathcal{X}$. The reference uniform distribution is always on the declared set; for a tuple, it is on the Cartesian product of the declared sets for its coordinates. Analogously, $\Delta$ can be defined as a distance between two *probability distributions* $\alpha, \beta : \mathcal{X} \to [0,1]$ as

$$\Delta(\alpha\,;\beta) := \Delta(A\,;B),$$

where $A$ and $B$ are random variables (taking values from $\mathcal{X}$) distributed according to $\alpha$ and $\beta$ (respectively). A similar convention applies to the distance $d$ from uniformity.

Randomized machines use private randomness independent of all external random choices.

**Exercise 1.1 An alternative definition of statistical distance**

Show that for every $A$ and $B$ we have

$$\Delta(A\,;B) = \sum_{a \in \mathcal{X}\colon \Pr[A=a] > \Pr[B=a]} \left(\Pr[A=a] - \Pr[B=a]\right).$$

**Exercise 1.2 Statistical distance as a metric**

Let $\Pi$ be the set of all probability distributions over a finite nonempty set $\mathcal{X}$. Prove that $\Delta$ is a *metric* on this set, i.e., it satisfies the following axioms:

- *non-negativity:* for every $\alpha, \beta \in \Pi$ we have $\Delta(\alpha\,;\beta) \geq 0$,
- *identity of indiscernibles:* for every $\alpha, \beta \in \Pi$ we have that $\Delta(\alpha\,;\beta) = 0$ if and only if $\alpha = \beta$,
- *symmetry:* for every $\alpha, \beta \in \Pi$ we have that $\Delta(\alpha\,;\beta) = \Delta(\beta\,;\alpha)$, and
- *triangle inequality:* for every $\alpha, \beta, \gamma \in \Pi$ we have that $\Delta(\alpha\,;\gamma) \leq \Delta(\alpha\,;\beta) + \Delta(\beta\,;\gamma)$.

**Exercise 1.3 Statistical distance as distinguishing advantage**

Let $A$ and $B$ be two random variables taking values from some finite set $\mathcal{X}$. For a (possibly randomized) machine $\mathcal{D}$ that takes inputs from $\mathcal{X}$ and outputs a bit, define its *distinguishing advantage* between $A$ and $B$ as

$$\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}}(A,B) := |\Pr[\mathcal{D}(A)=1] - \Pr[\mathcal{D}(B)=1]|.$$

[1] This is also known as the *total variation distance*, especially in the quantum information literature

Prove that for every computationally unbounded machine $\mathcal{D}$ we have

$$\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}}(A, B) \leq \Delta(A\,;B). \tag{1}$$

For every $A$ and $B$, construct $\mathcal{D}$ that achieves equality in Eq. (1). Show that the maximum is unchanged if the absolute value is omitted, and hence

$$\Delta(A\,;B) = \max_{\mathcal{D}} \left(\Pr[\mathcal{D}(A) = 1] - \Pr[\mathcal{D}(B) = 1]\right),$$

where the maximum is over all computationally unbounded machines that output a bit.

**Exercise 1.4 Statistical distance and guessing advantage**

Let $\alpha_0$ and $\alpha_1$ be two distributions over a finite nonempty set $\mathcal{X}$. Consider the following game played by a computationally unbounded, possibly randomized machine $\mathcal{M}$ (that knows $\alpha_0$ and $\alpha_1$):

1. A uniformly random bit $X \leftarrow \{0,1\}$ is chosen.
2. A value $A$ is sampled according to distribution $\alpha_X$ and sent to $\mathcal{M}$.
3. $\mathcal{M}$ receives $A$ and produces output $X' \in \{0,1\}$.

We say that $\mathcal{M}$ *won the game* if $X = X'$. Define its *guessing advantage* as

$$\mathsf{Adv}^{\mathrm{guess}}_{\mathcal{M}} := \left|\Pr[\mathcal{M} \text{ wins the game}] - \tfrac{1}{2}\right|.$$

Show that

$$\forall_{\mathcal{M}} \Pr[\mathcal{M} \text{ wins the game}] \leq \frac{1 + \Delta(\alpha_0\,;\alpha_1)}{2}. \tag{2}$$

For every $\alpha_0$ and $\alpha_1$, construct $\mathcal{M}$ that achieves equality in Eq. (2). By also considering the machine that complements its output, deduce that

$$\max_{\mathcal{M}} \mathsf{Adv}^{\mathrm{guess}}_{\mathcal{M}} = \tfrac{1}{2}\Delta(\alpha_0\,;\alpha_1).$$

**Exercise 1.5 How function application changes statistical distance**

Let $A$ and $B$ take values in a finite nonempty set $\mathcal{X}$, and let $\mathcal{Y}$ be a finite nonempty set. Show that for every function $f : \mathcal{X} \to \mathcal{Y}$ we have

$$\Delta(f(A)\,;f(B)) \leq \Delta(A\,;B).$$

Show that if $f$ is an injection, then

$$\Delta(f(A)\,;f(B)) = \Delta(A\,;B).$$

Deduce that if $f$ is a bijection from $\mathcal{X}$ onto $\mathcal{Y}$, then $d(f(A)) = d(A)$.

**Exercise 1.6 One-time pad with imperfect randomness**

Let $(\mathsf{Enc}, \mathsf{Dec})$ be the one-time pad encryption scheme for messages from $\{0,1\}^t$. Suppose a key $K$ is chosen from $\{0,1\}^t$ according to some distribution $\alpha$. Consider the following indistinguishability game played between a computationally unbounded adversary $\mathcal{A}$ and a challenger $\Omega$:

1. $\mathcal{A}$ produces two messages $m_0, m_1 \in \{0,1\}^t$ and sends them to $\Omega$.
2. $\Omega$ selects $X \leftarrow \{0,1\}$ uniformly at random, independently samples a fresh key $K$ according to $\alpha$, computes $c = \mathsf{Enc}(K, m_X)$, and sends $c$ to $\mathcal{A}$.
3. $\mathcal{A}$ receives $c$ and produces as output $X' \in \{0,1\}$.

We assume that $\mathcal{A}$ knows $\alpha$. We say that $\mathcal{A}$ *won the game* if $X = X'$. Show that

$$\forall_{\mathcal{A}} \Pr[\mathcal{A} \text{ wins the game}] \leq \tfrac{1}{2} + d(\alpha).$$

**Exercise 1.7 Conditional statistical distance**

Let $A$ and $B$ take values in finite nonempty sets $\mathcal{X}$ and $\mathcal{Y}$, respectively. For each $b$ with $\Pr[B = b] > 0$, let $\alpha_b$ be the conditional distribution of $A$ given $B = b$, viewed as a distribution over $\mathcal{X}$. Define the *statistical distance of $A$ from uniformity conditioned on $B$* as

$$d(A \mid B) := \sum_{b \in \mathcal{Y}:\, \Pr[B=b]>0} \Pr[B = b] \cdot d(\alpha_b).$$

Show that $d(A \mid B) = \Delta((A, B)\,; (U_{\mathcal{X}}, B))$, where $U_{\mathcal{X}}$ is uniform over $\mathcal{X}$ and independent of $B$.
Can you find a game-based interpretation of $d(A \mid B)$ similar to the one in Exercise Exercise 1.4?

**Exercise 1.8 Conditioning does not decrease the distance from uniformity**

Show that for every two random variables $A$ and $B$ taking values in finite nonempty sets, we have $d(A \mid B) \geq d(A)$.

## 2 Computational Indistinguishability

### Distribution ensembles

A *distribution ensemble* is a family of probability distributions

$$\mathcal{A} = \{A_n\}_{n \in \mathbb{N}},$$

indexed by a security parameter $n$. We assume that $A_n$ takes values in $\{0,1\}^{\ell(n)}$, where $\ell$ is polynomially bounded. As in the preceding section, we use the same notation for a distribution and a random variable distributed according to it.

Let $\mathcal{A} = \{A_n\}_{n \in \mathbb{N}}$ and $\mathcal{B} = \{B_n\}_{n \in \mathbb{N}}$ be two distribution ensembles over the same sets $\{0,1\}^{\ell(n)}$. For a probabilistic poly-time machine $\mathcal{D}$ that takes input $(1^n, x)$ and outputs a bit, define its *distinguishing advantage* as

$$\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}}(n) := |\Pr[\mathcal{D}(1^n, A_n) = 1] - \Pr[\mathcal{D}(1^n, B_n) = 1]|.$$

The probabilities are taken over the random variables $A_n$ and $B_n$, respectively, and the independent private randomness of $\mathcal{D}$.

A function $\mu : \mathbb{N} \to \mathbb{R}_{\geq 0}$ is *negligible* if for every constant $c > 0$ there exists $N_c \in \mathbb{N}$ such that for every $n \geq N_c$ we have $\mu(n) < n^{-c}$.

We say that $\mathcal{A}$ and $\mathcal{B}$ are *computationally indistinguishable*, denoted

$$\mathcal{A} \approx_c \mathcal{B},$$

if for every probabilistic poly-time machine $\mathcal{D}$ that outputs a bit, the function $\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}}$ is negligible.

### An equivalent formulation using samplers

A *sampler* is a probabilistic poly-time machine $\mathcal{S}$ that, on input $1^n$, outputs a string of length $\ell(n)$, where $\ell$ is polynomially bounded.

Let $\mathcal{S}_0$ and $\mathcal{S}_1$ be two samplers with the same output length $\ell(n)$. They induce the distribution ensembles

$$\mathcal{A} = \{\mathcal{S}_0(1^n)\}_{n \in \mathbb{N}} \quad \text{and} \quad \mathcal{B} = \{\mathcal{S}_1(1^n)\}_{n \in \mathbb{N}}.$$

We say that $\mathcal{S}_0$ and $\mathcal{S}_1$ are *computationally indistinguishable* if their induced distribution ensembles are computationally indistinguishable. Equivalently, for every probabilistic poly-time machine $\mathcal{D}$ that outputs a bit, the function

$$\varepsilon_{\mathcal{D}}(n) := |\Pr[\mathcal{D}(1^n, \mathcal{S}_0(1^n)) = 1] - \Pr[\mathcal{D}(1^n, \mathcal{S}_1(1^n)) = 1]|$$

is negligible. The probabilities are taken over the independent private randomness of the sampler and $\mathcal{D}$. This formulation applies to ensembles generated by probabilistic poly-time samplers; general distribution ensembles need not admit such samplers.

**Exercise 2.1 Guessing advantage and distinguishing advantage**

Let $\mathcal{A} = \{A_n\}_{n\in\mathbb{N}}$ and $\mathcal{B} = \{B_n\}_{n\in\mathbb{N}}$ be two distribution ensembles. Consider the following game played between a probabilistic poly-time adversary $\mathcal{M}$ and a challenger $\Omega$, with security parameter $n$:

1. $\Omega$ selects $X \leftarrow\!\!\$\ \{0,1\}$ uniformly at random.
2. If $X = 0$, $\Omega$ samples $x$ according to $A_n$. Otherwise, it samples $x$ according to $B_n$.
3. $\Omega$ sends $(1^n, x)$ to $\mathcal{M}$.
4. $\mathcal{M}$ receives $(1^n, x)$ and produces output $X' \in \{0,1\}$.

We say that $\mathcal{M}$ *won the game* if $X = X'$. Define its *guessing advantage* as

$$\mathsf{Adv}^{\mathrm{guess}}_{\mathcal{M}}(n) := \left|\Pr[\mathcal{M} \text{ wins the game}] - \tfrac{1}{2}\right|.$$

Show that

$$\mathsf{Adv}^{\mathrm{guess}}_{\mathcal{M}}(n) = \tfrac{1}{2}\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{M}}(n),$$

where $\mathcal{M}$ is also viewed as a machine that outputs a bit in the distinguishing experiment. Deduce that $\mathcal{A} \approx_c \mathcal{B}$ if and only if every probabilistic poly-time adversary has negligible guessing advantage.

**Exercise 2.2 Computational indistinguishability under efficient transformations**

Let $\mathcal{A} = \{A_n\}_{n\in\mathbb{N}}$ and $\mathcal{B} = \{B_n\}_{n\in\mathbb{N}}$ be two distribution ensembles over $\{0,1\}^{\ell(n)}$. Let $F$ be a probabilistic poly-time machine that, on input $(1^n, x)$ with $x \in \{0,1\}^{\ell(n)}$, outputs a string of length $m(n)$. Write $F_n(x)$ for the output of $F(1^n, x)$, including its private randomness.

(a) Suppose that $\mathcal{A} \approx_c \mathcal{B}$. Prove that

$$\{F_n(A_n)\}_{n\in\mathbb{N}} \approx_c \{F_n(B_n)\}_{n\in\mathbb{N}}.$$

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*Solution:* Suppose, towards a contradiction, that there exists a probabilistic poly-time machine $\mathcal{D}$ whose distinguishing advantage between the transformed ensembles,

$$\varepsilon(n) := |\Pr[\mathcal{D}(1^n, F_n(A_n)) = 1] - \Pr[\mathcal{D}(1^n, F_n(B_n)) = 1]|,$$

is non-negligible. Construct a machine $\mathcal{D}'$ as follows.

**Distinguisher $\mathcal{D}'$**

On input $(1^n, x)$ do the following:

1. Compute $y := F(1^n, x)$ using fresh private randomness.
2. Output $\mathcal{D}(1^n, y)$.

Since $F$ and $\mathcal{D}$ are probabilistic poly-time machines, so is $\mathcal{D}'$. Moreover,

$$\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}'}(n) = |\Pr[\mathcal{D}(1^n, F_n(A_n)) = 1] - \Pr[\mathcal{D}(1^n, F_n(B_n)) = 1]| = \varepsilon(n).$$

Thus $\mathcal{D}'$ distinguishes $\mathcal{A}$ from $\mathcal{B}$ with non-negligible advantage, contradicting $\mathcal{A} \approx_c \mathcal{B}$.

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2. Show that the converse of part (a) is false, i.e., that

$$\{F_n(A_n)\}_{n\in\mathbb{N}} \approx_c \{F_n(B_n)\}_{n\in\mathbb{N}}$$

does not necessarily imply $\mathcal{A} \approx_c \mathcal{B}$.

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*Solution:* Let $A_n = 0^n$ and $B_n = 1^n$ with probability 1. The poly-time machine $\mathcal{D}$ defined by

$$\mathcal{D}(1^n, x) := \begin{cases} 1 & \text{if } x = 0^n, \\ 0 & \text{otherwise} \end{cases}$$

has distinguishing advantage 1 for every $n \geq 1$. Hence $\mathcal{A} \not\approx_c \mathcal{B}$.

Now let $F(1^n, x) := 0^n$. This machine runs in polynomial time and satisfies $F_n(A_n) = 0^n = F_n(B_n)$. Thus the transformed distributions are identical for every $n$, and their ensembles are computationally indistinguishable.

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3. Suppose now that $F$ is deterministic and, for every $n$, the function

$$f_n : \{0,1\}^{\ell(n)} \to \{0,1\}^{m(n)}$$

computed by $F(1^n, \cdot)$ is injective. Does

$$\{f_n(A_n)\}_{n\in\mathbb{N}} \approx_c \{f_n(B_n)\}_{n\in\mathbb{N}}$$

necessarily imply $\mathcal{A} \approx_c \mathcal{B}$? Prove your answer or provide a counterexample.

## 3 Symmetric encryption

### Exercise 3.1 Negligible functions

Let $\varepsilon : \mathbb{N} \to \mathbb{R}$ be an arbitrary function. Split $\varepsilon$ into two functions: $\varepsilon_0$ and $\varepsilon_1$ that are equal to $\varepsilon$ on even and odd inputs, respectively. More precisely: for $b \in \{0,1\}$ let $\varepsilon_b$ be the function defined for every $m \in \mathbb{N}$ as $\varepsilon_b(m) := \varepsilon(2m + b)$. Show that if both $\varepsilon_0$ and $\varepsilon_1$ are negligible, then so is $\varepsilon$.

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*Solution:* To prove that $\varepsilon$ is negligible, take an arbitrary $c \in \mathbb{Z}$. We need to show that there exists $N \in \mathbb{N}$ such that for every $n > N$, we have that $|\varepsilon(n)| \leq n^{-c}$. Since for $\varepsilon_0$ and $\varepsilon_1$ are negligible, thus there exist $M_0$ and $M_1$ such that for every $m > M_b$ we have that $\varepsilon_b(m) \leq m^{-c-1}$. Hence, for every $m > \max(M_0, M_1)$ we have that

$$|\varepsilon_0(m)| \leq m^{-c-1} \text{ and } |\varepsilon_1(m)| \leq m^{-c-1},$$

which can be rewritten as

$$|\varepsilon(2m)| \leq m^{-c-1} \text{ and } |\varepsilon(2m+1)| \leq m^{-c-1}.$$

This implies that for every $n > \max(2M_0, 2M_1)$ we have $|\varepsilon(n)| \leq \lfloor n/2 \rfloor^{-c-1}$. Set $N := \max(2M_0, 2M_1, 3^{c+1})$. We get

$$\begin{aligned} |\varepsilon(n)| &\leq \lfloor n/2 \rfloor^{-c-1} \\ &\leq (n/3)^{-c-1} \qquad (3) \\ &\leq n^{-c} \cdot 3^{c+1}/n, \qquad (4) \end{aligned}$$

where in Eq. (3) we used the fact that $\lfloor n/2 \rfloor \geq n/3$ (which holds for every $n \geq 2$) and in Eq. (4) we used the fact that $n > N \geq 3^{c+1}$.

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### Exercise 3.2 Noticeable functions

A function $\mu$ is noticeable iff there exists $c \in \mathbb{N}$ and $n_0 \in \mathbb{N}$ such that for every $n \geq n_0$ we have that $|\mu(n)| \geq n^{-c}$. Answer the following questions:

(a) Is every noticeable function non-negligible?
(b) Is every non-negligible function noticeable?

**Exercise 3.3 One-way functions**

Let $f$ and $g$ be one-way functions that are *length-preserving*, i.e., for every $x \in \{0,1\}^*$ we have that $|f(x)| = |g(x)| = |x|$. For $h_i$'s defined as below, decide if they are one-way functions (if the answer is "yes" then give a proof, and otherwise provide a counterexample).

(a) $h_1(x) := f(x) \oplus g(x)$,
(b) $h_2(x) := f(g(x))$,
(c) $h_3(x) = f(x)||g(x)$,

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*Solution:* The answer is "no", i.e., it *is* possible that $f$ and $g$ are one-way, while $h_3$ is not. Below, we construct an example of such $f$ and $g$. Let $\alpha : \{0,1\}^* \to \{0,1\}^*$ be a length-preserving one-way function. We use the following notation: for a string $x = (x_1, \ldots, x_n)$ and $a, b \in \mathbb{N}$ (such that $1 \leq a \leq b \leq n$) let $x[a, \ldots, b]$ denote the string $(x_a, \ldots, x_b)$. For $x$ length $n$ define

$$f(x) := \alpha\left(x[1, \ldots, \lfloor n/2 \rfloor]\right) \,||\, x[\lfloor n/2 \rfloor + 1, \ldots, n]$$
$$g(x) := \alpha\left(x[\lfloor n/2 \rfloor + 1, \ldots, n]\right) \,||\, x[1, \ldots, \lfloor n/2 \rfloor].$$

For such $f$ and $g$ we have

$$h_3(x) = \alpha\left(x[1, \ldots, \lfloor n/2 \rfloor]\right) \,||\, x[\lfloor n/2 \rfloor + 1, \ldots, n] \,||\, \alpha\left(x[\lfloor n/2 \rfloor + 1, \ldots, n]\right) \,||\, x[1, \ldots, \lfloor n/2 \rfloor].$$

Since $x$ is a part of the output of $h(x)$, thus function $h_3$ is *not* one-way. It remains to show that $f$ and $g$ are one-way. Since these functions are symmetric, it suffices to prove this fact just for one of them, say $f$. We show that $f$ is one-way by contradiction. Suppose $f$ is not one-way, i.e., there exists a poly-time adversary $\mathcal{A}^f$ such that $\varepsilon$ defined as

$$\varepsilon(n) := \Pr\left[f(x') = y : x \leftarrow\!\$\ \{0,1\}^n;\ y := f(x);\ x' := \mathcal{A}^f(y)\right] \tag{5}$$

is non-negligible. For $b \in \{0,1\}$ let $\varepsilon_b$ be a function defined for every $m \in \mathbb{N}$ as $\varepsilon_b(m) := \varepsilon(2m + b)$. By Exercise 3.1 it cannot be the case that both $\varepsilon_0$ and $\varepsilon_1$ are negligible. Without loss of generality assume that $\varepsilon_0$ is non-negligible. We will construct an adversary $\mathcal{A}^\alpha$ that with non-negligible probability inverts $\alpha$. More precisely, $\mathcal{A}^\alpha$ will be such that

$$\Pr[\alpha(w') = v : w \leftarrow\!\$\ \{0,1\}^m;\ v := \alpha(w);\ w' := \mathcal{A}^\alpha(v)] = \varepsilon_0(m). \tag{6}$$

The adversary $\mathcal{A}^\alpha$ is presented below.

**Adversary $\mathcal{A}^\alpha$**

On input $v$ of length $m$ do the following:

1. sample $z \leftarrow\!\$\ \{0,1\}^m$,
2. let $q := \mathcal{A}^f(v \,||\, z)$,
3. output $q[1, \ldots, m]$.

It is easy to see that if $v := \alpha(w)$, where $w \leftarrow\!\$\ \{0,1\}^m$, then the distribution of the random variable $v \,||\, z$ in Step 2 of the above procedure is identical to the distribution of $y$ in the experiment from Eq. (5) with $n = 2m$. This means that the probability that $\mathcal{A}^f$ in Step 2 outputs $q$ such that $f(q) = v||z$ is equal to $\varepsilon(2m)$, which, in turn, is equal to $\varepsilon_0(n)$. By the definition of $f$ we have that $f(q) = v \,||\, z$ only if $\alpha(q[1, \ldots, m]) = v$. Hence Eq. (6) is proven. Note that the adversary $\mathcal{A}^\alpha$ just samples $z$ and runs $\mathcal{A}^f$ once. Since we assumed that $\mathcal{A}^f$ is poly-time, thus the running time of $\mathcal{A}^\alpha$ is also polynomial. Hence the proof that $f$ is one-way is finished. The proof for $g$ is analogous to the one for $f$.

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(d) $h_4(x) := x \oplus f(x)$.

**Exercise 3.4 Semantically-secure encryption implies that P≠NP**

Prove that if semantically secure encryption (with a key shorter than a message) exists, then P≠NP. *(for simplicity you can assume that the encryption scheme is deterministic)*

**Exercise 3.5 Semantically-secure encryption implies that P≠NP**

Prove that if one-way functions exist, then NP⊈BPP.

**Exercise 3.6 Time-memory trade-offs for inverting functions**

Let $H : \{0,1\}^n \to \{0,1\}^n$ be a function that is a bijection on $\{0,1\}^n$. Show an algorithm for inverting $H$ that runs in time $O(2^{n/2})$ if the following *preprocessing* is allowed: before obtaining $x$ the algorithm gets access to $H$ and fills in a table of size $O(2^{n/2})$ (assume that his time in this phase is unbounded).

*Bonus question:* what kind of a similar speed-up can one achieve without the assumption that $H$ is a bijection?

**Exercise 3.7 Composing PRGs**

Suppose $G : \{0,1\}^* \to \{0,1\}^*$ is a cryptographic pseudorandom generator and $\pi$ is a polynomial. Prove that $H : \{0,1\}^* \to \{0,1\}^*$ defined as a composition of two $G$'s, i.e.,

$$H(x) := G(G(x)),$$

is also a cryptographic pseudorandom generator. Think how to generalize this fact to the case when $H$ is a composition of a larger number of $G$'s.

*Hint:* Use the *hybrid argument*[2].

**Exercise 3.8 PRGs from hard-core bits**

Let $f : \{0,1\}^* \to \{0,1\}^*$ be a one-way function. We say that a poly-time computable function $\pi : \{0,1\}^* \to \{0,1\}$ is a *hard-core predicate for* $f$ if for every probabilistic poly-time adversary $\mathcal{A}$ we have that $\Pr[\mathcal{A}(f(x)) = \pi(x) : x \leftarrow\$ \{0,1\}^n] \leq 1/2 + \mathsf{negl}(n)$. Suppose $f$ is a one-way permutation, i.e., it is one-way and for every $n$ we have that $f$ restricted to $\{0,1\}^n$ is a permutation of this set. Define $G : \{0,1\}^* \to \{0,1\}^*$ as

$$G(x) = f(x) \,||\, \pi(x).$$

Show that $G$ is a pseudorandom generator.

**Exercise 3.9 PRG output extension**

Let $G$ be a pseudorandom generator that extends its input by one bit (i.e. it has expansion factor $\ell'$ such that $\ell'(n) = n + 1$). Let $\ell$ be any polynomial such that for every $n \in \mathbb{N}$ we have $\ell(n) > n$. Define $H : \{0,1\}^* \to \{0,1\}^*$ by the following procedure (for any $t \in \mathbb{N}$):

on input $(x_1^0, \ldots, x_t^0) \in \{0,1\}^t$:
**for** $i = 1, \ldots, \ell(t)$ **do**
  $(x_1^i, \ldots, x_{t+1}^i) := G(x_1^{i-1}, \ldots, x_t^{i-1})$
**return** $\left(x_{t+1}^1, \ldots, x_{t+1}^{\ell(t)}\right)$

Prove that $G$ is a pseudorandom generator.

[2] See, e.g., `https://en.wikipedia.org/wiki/Hybrid_{}argument_{}(Cryptography)`.

**Exercise 3.10 Two-Round Luby–Rackoff Is Not Secure.**

Let $f_1, f_2 : \{0,1\}^n \to \{0,1\}^n$ be arbitrary functions. Define a Feistel round by $T_f(L, R) = (R, L \oplus f(R))$. Consider the two-round Feistel permutation $P = T_{f_2} \circ T_{f_1}$ over $\{0,1\}^{2n}$.

Prove that $P$ can be efficiently distinguished from a uniformly random permutation using only two forward oracle queries, regardless of the choice of $f_1, f_2$.

**Exercise 3.11 Does Pointwise Uniformity Imply PRF Security?**

Let $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ be an efficiently computable keyed function family. Suppose that for every fixed $x \in \{0,1\}^n$, the random variable $F_K(x)$, where $K \leftarrow \{0,1\}^n$ uniformly, is uniformly distributed over $\{0,1\}^n$. Does this imply that $F$ is a secure PRF? Prove or disprove.

**Exercise 3.12 (Homework) PRF Security Implies Unpredictability.**

Let $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ be a secure pseudorandom function. Consider the following experiment:

1. Sample $K \leftarrow \{0,1\}^n$ uniformly.
2. Give a PPT adversary $\mathcal{A}$ oracle access to $F_K(\cdot)$, allowing polynomially many adaptive queries.
3. Eventually, $\mathcal{A}$ outputs $(x^*, y^*) \in \{0,1\}^n \times \{0,1\}^n$.
4. The adversary wins if $x^*$ was never queried to the oracle and $F_K(x^*) = y^*$.

Prove that for every PPT adversary $\mathcal{A}$,

$$\Pr[\mathcal{A} \text{ wins}] \leq 2^{-n} + \mathrm{negl}(n).$$

**Exercise 3.13 (Homework) From a PRF to a PRG.**

Let $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ be a secure PRF. For each $n \geq 2$, let $x_1, \ldots, x_n \in \{0,1\}^n$ be distinct, efficiently computable, publicly known strings. Define $G : \{0,1\}^n \to \{0,1\}^{n^2}$ by

$$G(k) = F_k(x_1) \| F_k(x_2) \| \cdots \| F_k(x_n).$$

Prove that $G$ is a secure pseudorandom generator.

**Exercise 3.14 A polynomially indexed sum of negligible functions**

For each $i \in \mathbb{N}$, let $\mu_i : \mathbb{N} \to \mathbb{R}_{\geq 0}$ be a negligible function. Show by example that, for a polynomially bounded function $t : \mathbb{N} \to \mathbb{N}$, the function

$$\mu(n) := \sum_{i=1}^{t(n)} \mu_i(n)$$

need not be negligible.

**Exercise 3.15 Polynomially many independent samples**

Let $\{X_n\}_{n \in \mathbb{N}}$ and $\{Y_n\}_{n \in \mathbb{N}}$ be computationally indistinguishable distribution ensembles, each efficiently samplable. For a polynomially bounded function $t : \mathbb{N} \to \mathbb{N}$, let $X_n^{\otimes t(n)}$ denote the joint distribution of $t(n)$ independent samples from $X_n$, and define $Y_n^{\otimes t(n)}$ analogously. Prove that

$$\{X_n^{\otimes t(n)}\}_{n \in \mathbb{N}} \approx_c \{Y_n^{\otimes t(n)}\}_{n \in \mathbb{N}}.$$

**Exercise 3.16 Random Functions vs. Random Permutations.**

Let $R : \{0,1\}^n \to \{0,1\}^n$ be a uniformly random function and let $\pi$ be a uniformly random permutation over $\{0,1\}^n$.

An adversary has forward oracle access to either $R$ or $\pi$ and may make at most $q \leq 2^n$ adaptive queries.

1. Construct an efficient distinguisher between these two oracles.

2. Prove that every distinguisher $D$, even a computationally unbounded one, making at most $q$ forward oracle queries satisfies
$$\left|\Pr[D^R = 1] - \Pr[D^\pi = 1]\right| \leq \frac{q(q-1)}{2^{n+1}}.$$
3. Deduce that every secure PRP family on $\{0,1\}^n$ is also a secure PRF family.

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*Solution:* Write $N := 2^n$ for the size of the domain and range. In the notation $D^O$, the distinguisher interacts with oracle $O$ and eventually outputs a bit. The probability $\Pr[D^R = 1]$ is over the choice of the random function and the private randomness of $D$; similarly, $\Pr[D^\pi = 1]$ is over the random permutation and the private randomness of $D$.

1. Query the oracle on $q$ distinct inputs $x_1, \ldots, x_q$, and output 1 if and only if two of the answers coincide. This procedure is efficient in $n$ and $q$ (and is polynomial-time when $q$ is polynomially bounded in $n$).
Let $Y_i$ be the answer to query $x_i$, and let $C$ be the event that $Y_i = Y_j$ for some $i < j$. Our distinguisher outputs 1 exactly when $C$ occurs. A permutation is injective, so $\Pr[D^\pi = 1] = 0$.
For a random function, $Y_1, \ldots, Y_q$ are independent and uniform over a set of size $N$. Conditional on the first $i$ answers being distinct, exactly $i$ outputs have already appeared. The next answer avoids these outputs with probability $(N-i)/N = 1 - i/N$. By multiplying these conditional probabilities, we obtain
$$\Pr[\text{all } q \text{ answers are distinct}] = \prod_{i=0}^{q-1} \left(1 - \frac{i}{N}\right).$$
Since $C$ is the complementary event, the distinguishing advantage is exactly
$$\Pr[D^R = 1] - \Pr[D^\pi = 1] = 1 - \prod_{i=0}^{q-1} \left(1 - \frac{i}{N}\right).$$
For $q = 0$ or $q = 1$, a collision is impossible and the advantage is zero (the empty product is 1).
One might instead output 1 whenever a collision occurs and, if no collision occurs, output 1 with probability $p$ and 0 with probability $1 - p$, using fresh randomness, where $0 \leq p \leq 1$. Does this improve the advantage? Let $c := \Pr[D^R = 1]$ be the collision probability computed above, and denote this modified algorithm by $D_p$. Splitting according to whether a collision occurs gives
$$\Pr[D_p^R = 1] = c + (1-c)p, \qquad \Pr[D_p^\pi = 1] = p.$$
The second equality holds because a permutation never produces a collision on distinct inputs. Therefore the modified distinguishing advantage is
$$\left|\Pr[D_p^R = 1] - \Pr[D_p^\pi = 1]\right| = |c + (1-c)p - p| = (1-p)c.$$
This is maximized at $p = 0$, which recovers our original algorithm. Thus randomizing after observing no collision gives no improvement; when $c > 0$, any $p > 0$ strictly reduces the advantage. This does not mean the original advantage is zero: it is $c$, which is negligible when $q$ is polynomially bounded in $n$.
To relate this to guessing the oracle, suppose the challenger chooses a random function or a random permutation with equal probability, and interpret output 1 as "random function" and output 0 as "random permutation." The original algorithm guesses correctly with probability
$$\frac{1}{2}\Pr[D^R = 1] + \frac{1}{2}\Pr[D^\pi = 0] = \frac{1}{2}c + \frac{1}{2} = \frac{1}{2} + \frac{c}{2}.$$
Thus a small probability of outputting 1 does not imply a small probability of guessing correctly: the algorithm always guesses correctly in the permutation experiment.

2. We prove the bound using statistical distance. The main idea is to compare a list of independent uniform answers with a list of uniform answers chosen without repetition.

**Step 1: Compare the two answer lists.** Let $A = (A_1, \ldots, A_q)$ consist of $q$ independent uniform elements of $\{0,1\}^n$. Let $P = (P_1, \ldots, P_q)$ be sampled uniformly from all ordered lists of $q$ distinct elements of $\{0,1\}^n$. These correspond to sampling with replacement and without replacement, respectively.
Let $C$ be the set of lists containing a repeated entry, and write $c := \Pr[A \in C]$. The list $P$ never belongs to $C$. Moreover, conditioned on $A \notin C$, the list $A$ is uniform among all lists of distinct entries, so it has exactly the distribution of $P$.
We can now compute their statistical distance directly. For a list $a \in C$, $\Pr[P = a] = 0$. For a list $a \notin C$,

$$\Pr[A = a] = (1 - c)\Pr[P = a].$$

Therefore,

$$\begin{aligned}\Delta(A\,;P) &= \frac{1}{2}\sum_{a\in C}\Pr[A=a] + \frac{1}{2}\sum_{a\notin C}\big(\Pr[P=a] - \Pr[A=a]\big)\\ &= \frac{1}{2}c + \frac{1}{2}\big(1-(1-c)\big) = c.\end{aligned}$$

Thus the statistical distance is exactly the probability of a repetition in the independent list.

**Step 2: Bound the repetition probability.** For each pair $1 \le i < j \le q$, define the event

$$E_{i,j} := \{A_i = A_j\}.$$

Since $A_i$ and $A_j$ are independent and uniform, $\Pr[E_{i,j}] = 1/N$: after fixing $A_i$, exactly one of the $N$ possible values of $A_j$ matches it. A repetition occurs if at least one of these events occurs. The union bound therefore gives

$$c = \Pr\left[\bigcup_{i<j} E_{i,j}\right] \le \sum_{i<j}\Pr[E_{i,j}] = \binom{q}{2}\frac{1}{N} = \frac{q(q-1)}{2^{n+1}}.$$

There are $\binom{q}{2}$ pairs. The events need not be independent for the union bound to apply.

**Step 3: Use the lists to answer adaptive queries.** To connect these answer lists to the oracle experiment, imagine choosing a list $(z_1, \ldots, z_q)$ before running $D$. Answer the first query with $z_1$. When $D$ queries a different input for the first time, answer with $z_2$, and so on: the $j$th distinct input receives $z_j$. Record the answer assigned to each input. If $D$ queries that input again, return the same recorded answer, rather than taking another entry from the list. For example, queries $x, x, x'$ with $x' \neq x$ receive answers $z_1, z_1, z_2$. Since $D$ makes at most $q$ queries, the list always contains enough entries.
Using $A$ simulates a random function: every new input receives an independent uniform answer. Using $P$ simulates a random permutation: every new input receives a uniform answer among outputs not used before. This remains true for adaptive queries. Once the previous queries and answers are fixed, the value at a new input has precisely the stated distribution in each experiment.
Fix the private randomness of $D$ to some value $r$. The simulation now turns an answer list into a final output bit by a deterministic function, say $g_r$. Statistical distance cannot increase under a deterministic function. Also, for bit-valued random variables, statistical distance equals the absolute difference between their probabilities of being 1. Hence

$$|\Pr[g_r(A) = 1] - \Pr[g_r(P) = 1]| \le \Delta(A\,;P) \le \frac{q(q-1)}{2^{n+1}}.$$

This bound holds for every fixed $r$. Averaging over the private randomness of $D$, and using that the absolute value of an average is at most the average of the absolute values, yields

$$\big|\Pr[D^R = 1] - \Pr[D^\pi = 1]\big| \le \frac{q(q-1)}{2^{n+1}}.$$

The proof only limits the number of queries. It places no restriction on the computation performed by $D$, so the result also holds for computationally unbounded distinguishers.

3. Let $F$ be a secure PRP family, and let $D$ be any probabilistic polynomial-time PRF distinguisher making at most $q(n)$ queries, where $q$ is polynomially bounded. Here $K$ is a uniformly sampled key, and all experiments also include the private randomness of $D$. We compare three experiments: the oracle is $F_K$, a random permutation $\pi$, or a random function $R$. Insert the random-permutation experiment between the other two and apply the triangle inequality, followed by part 2:

$$\begin{aligned}\left|\Pr[D^{F_K}=1]-\Pr[D^{R}=1]\right| &\le \left|\Pr[D^{F_K}=1]-\Pr[D^{\pi}=1]\right| + \left|\Pr[D^{\pi}=1]-\Pr[D^{R}=1]\right| \\ &\le \left|\Pr[D^{F_K}=1]-\Pr[D^{\pi}=1]\right| + \frac{q(n)(q(n)-1)}{2^{n+1}}.\end{aligned}$$

The first term is negligible by PRP security: a PRP distinguisher can run $D$, forward all its queries to its own oracle, and return its output bit. This simulation is exact and runs in polynomial time. This is where efficiency of $D$ is needed.
For the second term, choose constants $C, d > 0$ such that $q(n) \le Cn^d$ for all sufficiently large $n$. Then

$$\frac{q(n)(q(n)-1)}{2^{n+1}} \le \frac{C^2 n^{2d}}{2^{n+1}}.$$

For every constant $c > 0$, exponential growth eventually gives $2^{n+1} > C^2 n^{2d+c}$, so the last expression is smaller than $n^{-c}$. Hence this term is negligible. Also, $q(n) \le 2^n$ for sufficiently large $n$, which justifies applying part 2; finitely many smaller security parameters do not affect negligibility.
The sum of these two negligible functions is negligible. Since this holds for every probabilistic polynomial-time $D$, the family $F$ is a secure PRF family.

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**Exercise 3.17 IND-CPA-Secure Encryption from PRFs/PRPs.**

Let

$$F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$$

be a secure pseudorandom function.

Consider the following symmetric encryption scheme:

- **Key generation:** Sample $K \leftarrow \{0,1\}^n$ uniformly.
- **Encryption:** On input $m \in \{0,1\}^n$, sample $r \leftarrow \{0,1\}^n$ uniformly and output

$$\mathrm{Enc}_K(m) = (r, F_K(r) \oplus m).$$

- **Decryption:** On input $(r, z)$, output

$$\mathrm{Dec}_K(r, z) = F_K(r) \oplus z.$$

1. Prove that the scheme is correct.
2. Prove that the scheme is IND-CPA secure if $F$ is a secure PRF.

**Exercise 3.18 (homework) CBC Security and Unpredictable Initialization**

Let $F$ be a secure pseudorandom permutation on $\{0,1\}^n$. For a message $M = (m_1, \ldots, m_\ell)$ consisting of $\ell \ge 1$ blocks in $\{0,1\}^n$, define CBC encryption by sampling a fresh, independent uniform initialization vector $c_0 \leftarrow\!\$\, \{0,1\}^n$ and setting

$$c_i = F_K(m_i \oplus c_{i-1}), \qquad i = 1, \ldots, \ell.$$

The ciphertext is $(c_0, \ldots, c_\ell)$. Each message is submitted in full before its initialization vector is sampled; the two challenge messages must have the same number of blocks.

Prove IND-CPA security against adaptive probabilistic polynomial-time adversaries. Specifically, let $\mathcal{A}$ be an adversary whose encryption queries (before and after the challenge), together with the message encrypted in the challenge, contain at most $\sigma$ plaintext blocks altogether. Construct a PRP distinguisher $\mathcal{B}$ making at most $\sigma$ forward oracle queries and prove

$$\mathsf{Adv}^{\text{IND-CPA}}_{\text{CBC}}(\mathcal{A}) \leq \mathsf{Adv}^{\text{PRP}}_{F}(\mathcal{B}) + O\left(\frac{\sigma^2}{2^n}\right).$$

Here IND-CPA advantage is $\left|\Pr[b' = b] - \frac{1}{2}\right|$, and PRP advantage is the absolute difference between $\mathcal{B}$'s probabilities of outputting 1 with oracle $F_K$ and with a uniformly random permutation. Your proof must identify the relevant collision event and justify its probability despite adaptive message selection.

**Exercise 3.19 (homework) Single-Bit to Multi-Bit IND-CPA Encryption.**

Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be an IND-CPA-secure symmetric encryption scheme with message space $\{0,1\}$. Assume that encryption is stateless and randomized, using fresh independent randomness on every invocation.

Let $\ell(n) \geq 1$ be polynomially bounded. Construct a scheme $\Pi^\ell$ encrypting messages from $\{0,1\}^{\ell(n)}$ by independently encrypting each bit under the same secret key.

Prove that $\Pi^\ell$ is IND-CPA secure. In particular, show that every PPT adversary breaking $\Pi^\ell$ with advantage $\varepsilon(n)$ yields a PPT adversary breaking $\Pi$ with advantage at least $\frac{\varepsilon(n)}{\ell(n)}$. Here IND-CPA advantage is defined as $\left|\Pr[b' = b] - \frac{1}{2}\right|$.

**Exercise 3.20 (homework) IND-CPA-Secure Encryption Combiner.**

Let $\Pi_1 = (\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ and $\Pi_2 = (\text{Gen}_2, \text{Enc}_2, \text{Dec}_2)$ be two correct, efficiently computable symmetric encryption schemes supporting all binary messages of polynomially bounded length. Construct a symmetric encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ satisfying the following property:

$$\begin{array}{l} \Pi_1 \text{ is IND-CPA secure} \\ \qquad \text{or} \qquad\quad \Longrightarrow \quad \Pi \text{ is IND-CPA secure.} \\ \Pi_2 \text{ is IND-CPA secure} \end{array}$$

Assume that the keys are generated independently. Prove correctness and IND-CPA security of your construction.